Code No: 136FC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
FUNDAMENTALS OF ROBOTICS
(Common to CE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used Paper Aug-2024

PART – A

(25 Marks)

1.a)	Outline the introduction of robotics.	[2]
b)	Define accuracy and repeatability.	[3]
c)	Label the mechanisms for actuation.	[2]
d)	List the considerations in gripper selection.	[3]
e)	Define kinematics.	[2]
f)	What is the significance of homogenous transformation matrix?	[3]
g)	Label steps involved in trajectory planning.	[2]
h)	Outline computed torque control.	[3]
i)	List the applications of robot vision.	[2]
j)	What is the significance of Lead through programming method?	[3]

PART – B

(50 Marks)

2. Explain the classification of robotics with respect to geometrical configuration (Anatomy) and controlled system. [10]

OR

3. Analyze position sensors and velocity sensors in detail. [10]

4. Summarize the working of mechanical Gripper. [10]

OR

5. Explain operating principle of magnetic gripper. [10]

6. Compare direct and inverse kinematics for industrial robots. [10]

OR

7. Evaluate differential kinematics for planar serial robots. [10]

8. Analyze interaction control and rigid body mechanics. [10]

OR

9. Explain control architecture with position, path velocity, and force control systems. [10]

10. Describe the programming of robots and vision System. [10]

OR

11. Explain Teach pendent and textual programming language of VAL in detail. [10]

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